Competitive Landscape: Consumption-Based Model for On-Premises Infrastructure

5 March 2024 - ID G00771490 - 18 min read

By: Kiyomi Yamada, Adrian O'Connell, Jeff Vogel, Tony Harvey

Initiatives:Technology Market Essentials; Capitalize on the Tech Market Landscape

The consumption-based model for on-premises is becoming one of the key strategies for data center infrastructure providers as alternatives to traditional IT sales. They must understand the current trends and positions of other competitors.

Overview

Key Findings

- Providers use the consumption-based model (CBM) to provide a cloudlike payment model with flexible pay-for-use or -access financing options as an alternative to their traditional commercial/payment modes, requiring large capex commitments from their customers.
- CBM potentially produces recurring revenue streams from subscription business models while gaining frequent customer engagement opportunities, helping providers to increase revenue and customer lifetime value.
- CBM programs basically bundle services and hardware/software solutions, aiming to reduce administrative overhead for life cycle management and maintenance.

Recommendations

- Prioritize the existing customers that have a large-scale installed base by motivating them to switch current hardware with CBM contracts.
- Provide channel partners an opportunity to grow their solution business by combining with a more flexible CBM pay-as-you-grow model while simplifying and automating the deal process to make it easier for them.
- Offer fully managed services to address operational efficiency and internal IT resource shortage, as well as vertical solutions, by partnering with third-party independent software vendors (ISVs).

This research note is restricted to the personal use of GERMAN.AGUNDEZ@IKUSI.COM.

Analysis

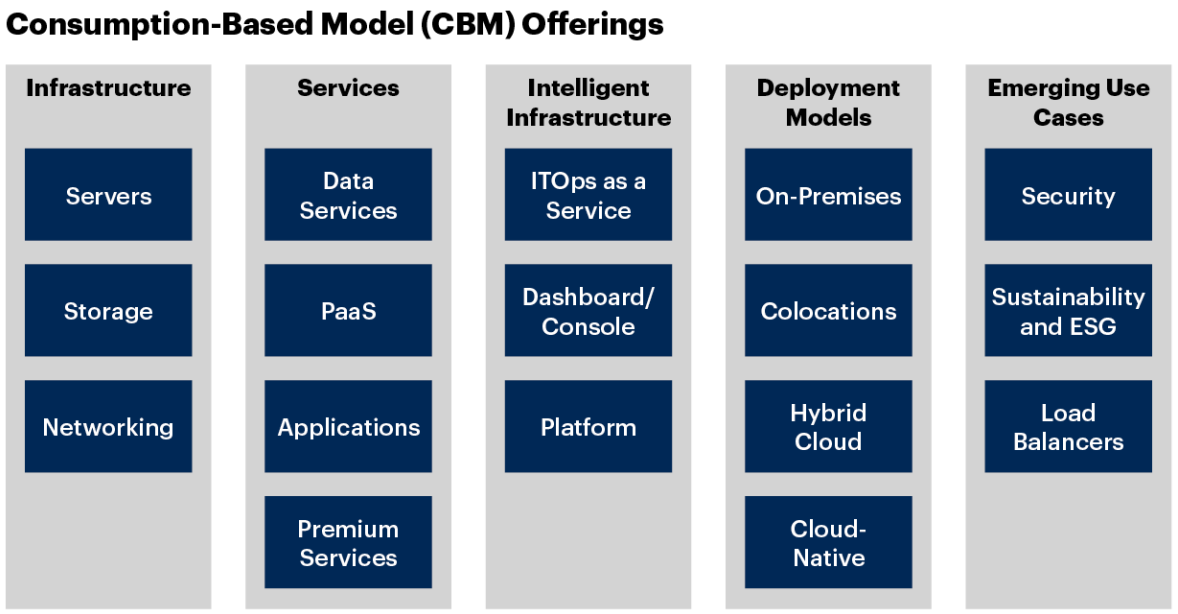
Introduction

The consumption-based model (CBM) for on-premises storage, compute and networking data center infrastructure is an alternative on- and off-premises offering, enabling payments that are simpler, easier, and more reflective of their actual usage, compared to traditional hardware purchases. CBM has been becoming one of the key strategies for many providers as their customers are demanding simplified and agile data center infrastructure deployments. Many IT operations are increasingly getting used to the cloud model where they pay only for what is used or consumed. Financial motivations such as reducing capital expenditure (capex) and predictable usage fees in alignment with optimized asset utilization are also driving interest in CBM.

Figure 1 details current CBM offerings, including all data center infrastructure; servers, storage and networking, although networking CBM is still nascent. In addition to infrastructure, they provide services running on intelligent infrastructure, such as IT operations as a service. As to locations, typically users can choose on-premises, colocation or partners'/vendors' sites. Some infrastructure offers hybrid and/or multicloud options, including use of cloud-native platforms. Also, emerging use cases like ransomware protection, security and sustainability are often packaged under some CBM programs.

This research note is restricted to the personal use of GERMAN.AGUNDEZ@IKUSI.COM.

Figure 1: Consumption-Based Model (CBM) Offerings



Source: Gartner
ESG = environmental, social and governance; PaaS = platform as a service
771490_C

Gartner

CBM History and Present

In early days, many CBM programs targeted computing. However, with increasing data volumes, more storage programs have emerged to help users minimize overprovisioning and increase flexibility. They have also been used to test new products (such as hyperconverged infrastructure) or providers. Many CBM programs focused on the “pay per use” aspect, although they were not truly “consumption based” with a minimal term and/or capacity requirement along with a lengthy contract process (pay for use [PFU]). For storage, the subscription (pay for access [PFA]) type has been most prevalent, but we have started seeing more PFU programs.

This research note is restricted to the personal use of GERMAN.AGUNDEZ@IKUSI.COM.

The consumption-based model (CBM) for data center infrastructure is an as-a-service offering based on pay for use (PFU) or pay for access (PFA) as an alternative to the traditional hardware purchase model (buying or leasing). CBM is inspired by cloud service adoption with flexible payment options (capex or operating expenditure [opex]). Most programs have a minimum term and/or capacity requirement.

Two Types of CBM — Usage (PFU) Versus Subscription (PFA)

CBM includes both usage-based (PFU) and subscription access fee plans (PFA) that vary by license and contract terms. CBM PFU primarily offers service-level agreements (SLAs) based on usage, while PFA offers access to platform services under a subscription. Each type has its own benefits and drawbacks. See Figure 1 from Market Guide for Consumption-Based Models for Data Center Infrastructure.

CBM — Usage (PFU)

Under the PFU CBM, users and vendors first negotiate metrics-based SLA requirements, usage-based commitment levels and per-unit costs. Following this step, the client collaborates with the vendor on which tier of services — performance, workload, costs, etc. — is best suited to deliver SLA requirements. Users can choose between managing the infrastructure or having the vendor or partner manage the infrastructure under the SLA. Options in terms of transaction financing vary by vendor, with some offering portions of the contract term paid upfront to drive discounts and then expensed monthly, based on the metered amount used.

Program examples:

- Dell Technologies APEX Flex on Demand
- Hewlett Packard Enterprise (HPE) GreenLake for Private Cloud Enterprise
- Lenovo TruScale Infinite Storage
- NetApp Keystone STaaS (PFU version recently launched)
- Pure Storage Evergreen//One

This research note is restricted to the personal use of GERMAN.AGUNDEZ@IKUSI.COM.

CBM — Subscription (PFA)

CBM PFA offers access to platform services under a subscription. PFA value is intrinsically associated with access to platform services, or tiered offerings based on price and type of service. Under a PFA license, usage isn't billed directly as it's consumed. Instead, access to the service provided is billed. Users can expect the vendor to maintain the equipment through maintenance and technology refresh cycles.

Program examples:

- Cisco+ Secure Connect
- Dell APEX Compute and Data Services
- HPE GreenLake for Aruba
- Lenovo TruScale Infrastructure as a Service (Fixed Option)
- NetApp Keystone STaaS (subscription)
- Pure Storage Evergreen//Forever and Evergreen//Flex

Competitive Situation and Trends

The CBM is becoming more common with increased offerings and promotions. Client education is still needed as these programs are not easy to understand. Most of the programs are still evolving, and many variances exist in terms of contract process and covered products.

The providers in this Competitive Landscape are depicted based on how they focus their CBM strategies. Some providers have an extensive portfolio, while some focus on the selected markets. They also differ in how deep they go into the infrastructure stack with their CBM offerings.

Data center infrastructure providers have been making a significant effort to expand their CBM programs as a vehicle to increase customer retention. Providers aim to:

- Limit the pace of public cloud migrations
- Expand customer lifetime value
- Maximize attachments for software and services

This research note is restricted to the personal use of GERMAN.AGUNDEZ@IKUSI.COM.

- Attain higher renewal rates for the existing customer base

Hedging the pace of cloud migrations — For various reasons such as latency issues, regulations, privacy and security, as well as available internal IT skill sets and cultural factors, many users opt for on-premises (including colocation) deployments. Providers use CBM to provide a cloudlike payment model with flexible financing options compared to their traditional offerings requiring large capex requests. Public cloud vendors have simultaneously introduced on-premises products as extensions to the public cloud and as part of a central IT operations control console for onboarding services (e.g., Amazon Web Services [AWS] Outposts and Microsoft Azure Stack Hub provide on-premises instantiations).

Extending customer lifetime value — CBM enables providers to produce predictive recurring revenue streams from their recurrent revenue business models. CBM also brings more frequent customer engagement opportunities that potentially help providers to increase revenue and customer lifetime value.

Increasing attachment rates — Providers try to maximize their software and service attachments that wrap around their hardware offerings because software and services generally provide better margins. Often, CBM programs include the use of artificial intelligence for IT operations (AIOps) tools, as well as features and capabilities that augment on-premises administration and support activities, thereby lowering overall IT administration costs. Ultimately providers are trying to move up the stack from infrastructure to applications (see Figure 2).

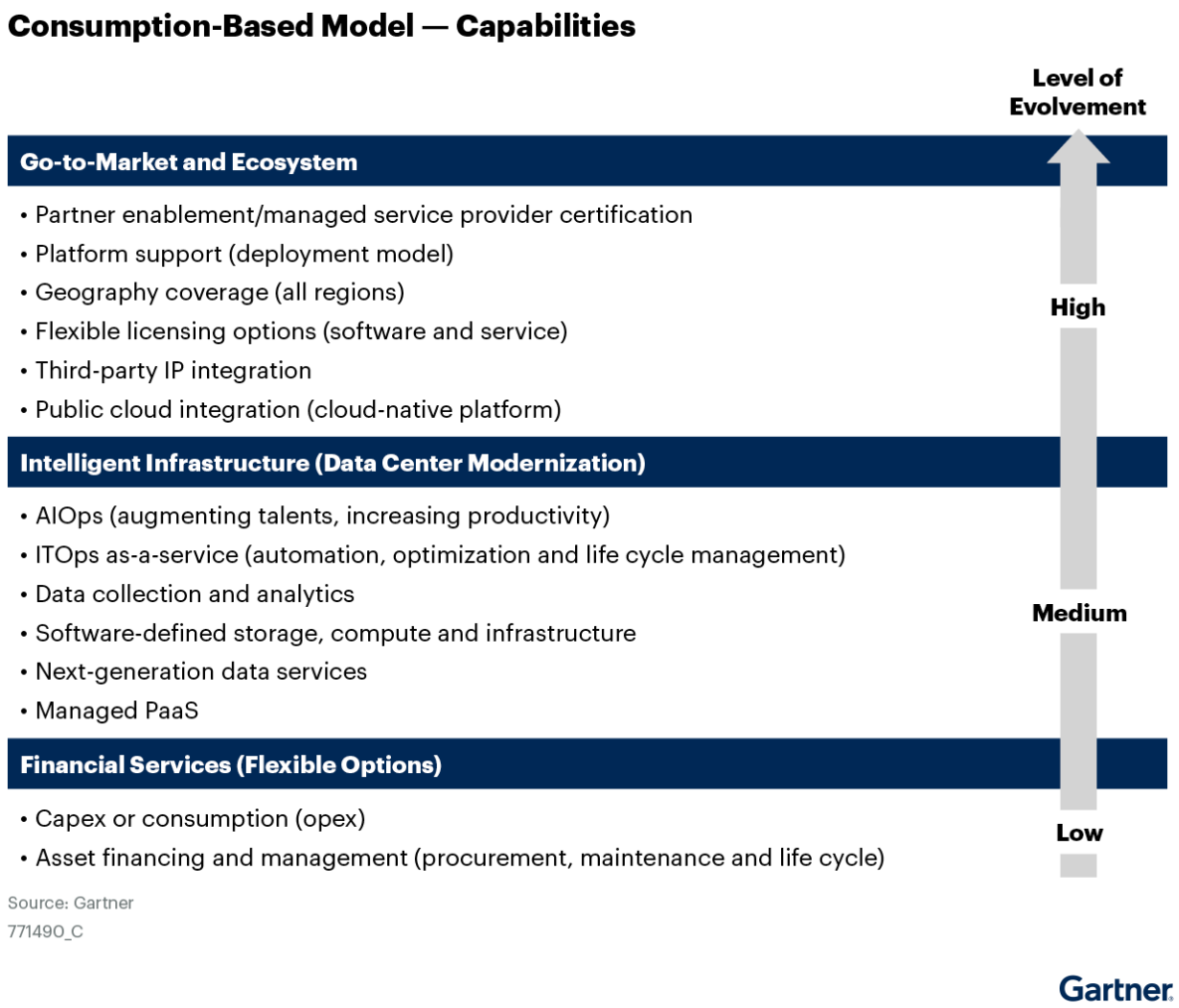
Attaining higher renewal rates for the existing customer base — Gartner believes CBM customers have higher rates of renewal compared with traditional buying because customers see the value of risk sharing with their providers. Some like freeing up their duty to manage and maintain their data center equipment as well as technology refresh cycles. Others prefer the flexibility that CBM programs provide.

Providers Trying to Move Up in the Stack With Their CBM Programs

Figure 2 details the capabilities that CBM programs could offer. There are differences in terms of these capabilities among the CBM providers, but ideally they should aim to hit all items listed here.

This research note is restricted to the personal use of GERMAN.AGUNDEZ@IKUSI.COM.

Figure 2: Consumption-Based Model — Capabilities



Of all capabilities, IT/AIOps capability (including console) is critical because it is the center of CBM service offerings, providing automation or optimization while providers can offer differentiation. Table 1 summarizes the capabilities by the representative providers.

This research note is restricted to the personal use of GERMAN.AGUNDEZ@IKUSI.COM.

Table 1: CBM IT/AI Capabilities by Vendor

(Enlarged table in Appendix)

Vendor	Console	AIOps (integrated in console) — management, analytics and cloud ops management
Cisco	Cisco Intersight and Cisco Nexus Dashboard	Cisco Intersight Infrastructure Service and Cisco Intersight Workload Optimizer
Dell Technologies	APEX Console	APEX Console, CloudIQ and others
Hewlett Packard Enterprise (HPE)	HPE GreenLake edge-to-cloud platform	HPE InfoSight and Aruba Central
Hitachi Vantara	Hitachi Ops Center	Hitachi proprietary and third-party tools
IBM Power	IBM Cloud Management Console, Hardware Management Console	IBM proprietary tools
IBM Storage	IBM Storage Insights	IBM proprietary tools
Lenovo	XClarity	XClarity, NetApp ONTAP and third party
NetApp	BlueXP	Active IQ and Cloud Insights
Pure Storage	Pure1	Pure1

Source: Gartner (November 2022)

This research note is restricted to the personal use of GERMAN.AGUNDEZ@IKUSI.COM.

Competitive Profiles

Cisco

Product or Portfolio Overview

In 2021, Cisco launched a new as-a-service brand, Cisco+, which provides customers with consumption-based customer experience. The first offering is Cisco+ Hybrid Cloud (bare-metal, compute, edge, VDI, virtualization and data center networking). Cisco+ Hybrid Cloud is a solution package, including networking, compute and storage hardware, software, and services for on-premises with an option of self-managed or full-managed services from our Cisco+ partners. Cisco uses in-house platform suites, including Intersight (hybrid cloud operation platform), delivering automation, observability and optimization (in conjunction with other management tools, such as Cisco Intersight Workload Optimizer, Cisco Nexus Dashboard and Cisco ACI) to manage networking products. Cisco+ Hybrid Cloud is available in six countries as of September 2022 and the company reported its expansion in the future.

How This Provider Competes

Cisco has longtime CBM experiences from Cisco Open Pay from Cisco Capital although it was mainly offered as financing alternatives. For Cisco+, Cisco utilizes its own products (hardware such as HyperFlex, UCS servers and Nexus, software and APIs for hybrid cloud offerings) and plans to add more portfolios, such as security and networking, as well as incremental enhancements in Cisco+. It started offering a new cloud-based Secure Access Service Edge (SASE) solution known as Cisco+ Secure Connect Now that combines Cisco Meraki SD-WAN with Cisco's work-from-anywhere security solutions through an integrated platform. The company can pave the way for networking as a service as one of the leaders in this space.

This research note is restricted to the personal use of GERMAN.AGUNDEZ@IKUSI.COM.

Dell Technologies

Product or Portfolio Overview

Dell Technologies provides extensive as-a-service portfolios under its APEX brand, including the following:

APEX Data Storage Services, APEX Cloud Services with VMware Cloud, APEX Private and Hybrid Cloud, APEX Backup Services, APEX Cyber Recovery Services, and custom solutions including APEX Flex on Demand and APEX Data Center Utility. Additionally, through Dell's partnership with Equinix, Dell APEX is available in a Dell-managed, interconnected colocation facility. This enables customers to take advantage of public cloud resources. Dell APEX is available in 39 countries globally (although some services are only offered in a subset of these countries) through both direct and partners as of September 2022.

How This Provider Competes

Dell also had decades of experience for custom CBM solutions before launching its APEX brand. APEX leverages Dell's wide range of products at on-premises or in colocation facilities, providing the benefits of the cloud experience. The APEX Console provides a single user interface (UI) for various AIOps tasks including infrastructure management, analytics and cloud management. For example, with the APEX Console, users will be able to order and manage multicloud connectivity and dynamically allocate bandwidth to various public clouds, as needed. They can choose from a variety of compute instance types that match their workloads, number of instances, and the amount of storage and place an order.

Hewlett Packard Enterprise (HPE)

Product or Portfolio Overview

Under the HPE GreenLake umbrella brand, HPE has a stated goal of moving its product portfolio to be everything as a service (XaaS) and pay-per-use pricing by the end of 2022. HPE GreenLake has since expanded the program to include software and data services, as well as deployments on-premises and in colocation facilities. HPE GreenLake edge-to-cloud platform unifies the management of your IT environment at all levels from devices to workloads, including analytics and AIOps capabilities. HPE GreenLake service assigns a technical support individual to the account who performs the capacity management and planned changes and coordinates additional capacity as needed. HPE GreenLake offerings have been available in over 50 countries as of September 2022.

This research note is restricted to the personal use of GERMAN.AGUNDEZ@IKUSI.COM.

How This Provider Competes

As one of the CBM pioneers, HPE advanced HPE GreenLake's strategy with a portfolio of integrated products, services and partner-based workload solutions. HPE has continued to extend its GreenLake portfolio offerings, ranging from silicon-to-applications to XaaS infrastructure. It expanded its GreenLake offerings with an update to its storage portfolio with enhanced versions of its primary storage products with automated provisioning and nondisruptive upgrades. A catalog of software and partner-centric workload applications provides customers an integrated solution experience that removes complexity and IT operational risks. Industry verticalization of advisory services with major solution partners accelerates time-to-solution value within key vertical industries.

Hitachi Vantara

Product or Portfolio Overview

EverFlex from Hitachi is the portfolio of products, services and solutions that Hitachi Vantara delivers as a service. The EverFlex portfolio includes storage as a service (STaaS) from Hitachi Vantara and data protection as a service (DPaaS) from Hitachi Vantara for backup and recovery with remote replication as an option. Also included are compute and infrastructure resources as a service for both VMware and cloud-native application environments, as part of the custom offerings delivered by Hitachi Vantara managed services. The EverFlex portfolio of as-a-service offerings includes operations management delivered as a service (Hitachi Infrastructure Orchestration as a Service), Kubernetes management as a service, and the as-a-service delivery of ERP application-run services through Hitachi Vantara and oXya. The EverFlex offerings have been available in 28 countries through both the vendor and partners as of September 2022.

How This Provider Competes

Hitachi Vantara provides on-demand access to computing power, along with modern data and application services for developers in on-premises, private cloud environments. Admins can include additional managed services for capacity monitoring, reporting, capacity management and provisioning, security management, infrastructure management, usage dashboards and a self-service portal. The company is expanding its EverFlex portfolio to include workload management across hybrid cloud environments, as well as run application reliability operations on public cloud resources managed as a service. The newly added infrastructure as a service has enhanced its CBM portfolio by offering autonomous service-based IT operations.

This research note is restricted to the personal use of GERMAN.AGUNDEZ@IKUSI.COM.

IBM

Product or Portfolio Overview

IBM has had programs with CBM elements in both its server and storage portfolios for many years. It has offered capacity on demand for its IBM Power servers (Power Private Cloud with Dynamic Capacity and options including IBM Elastic Capacity on Demand or Shared Utility Capacity across a collection of IBM Power servers). IBM has also offered capacity on demand for IBM Z mainframe servers (IBM Z Tailored Fit Pricing — mainframe software and hardware on a consumption basis), in which users can activate additional dormant processing capacity on purchased hardware. IBM Storage offers two CBM options: One is an aaS model with a set of predefined service levels to choose from, and the other is based on deploying custom configurations tailored to specific client needs. Both have a base subscription/capacity level with dynamic, pay-as-you-use (up or down) consumption model that shifts traditional purchase capacity over provisioning risk. IBM Power server CBM offerings are available in 67 countries and its storage programs in 37 countries as of September 2022.

How This Provider Competes

IBM's CBM programs attract users with fixed capex budgets with unpredictable future capacity requirements. A portion of the hardware's capacity is activated at purchase, and the additional capacity can be activated later. Elastic Capacity on Demand (temporary activation of processor or memory resources) can be charged by the day, and Shared Utility Capacity (system resource entitlements shared across a pool of metered capacity) can be metered by the minute. IBM Power also enabled Red Hat OpenShift Container Platform as a base/metered capacity element in a Power Enterprise Pool. IBM Block Storage as a Service leverages IBM Spectrum Virtualize software that extends the service for hybrid cloud use cases on public cloud infrastructure and provides IBM concierge service (via a technical account manager assigned to each client). The IBM Storage Utility offering provides a custom CBM storage option providing a flexible consumption across most of IBM's storage portfolio (IBM's Block, File, Object and Virtual Tape families).

This research note is restricted to the personal use of GERMAN.AGUNDEZ@IKUSI.COM.

Lenovo

Product or Portfolio Overview

Lenovo TruScale offers a broad portfolio of as-a-service solutions for its line of products, including end-user devices, servers, storage and integrated systems. TruScale IaaS solutions include ThinkSystem and ThinkAgile HCI systems integrated with VMware vSAN, VMware Cloud Foundation, Microsoft Azure Stack or Nutanix HCI software, and an SAP S/4HANA Cloud partnered solution. These are in addition to the original offering for VDI with Nutanix and Citrix. New additions include TruScale Infinite Storage, TruScale Backup Services and TruScale for High Performance Computing (HPC). Third-party software and other enterprise components can be bundled as part of a TruScale solution. TruScale can be deployed globally in any size, with flexible capacity commitments. TruScale also includes device-as-a-service and SaaS offerings.

How This Provider Competes

As a relatively new entry for this space, Lenovo has been catching up to extend its CBM offerings based on its service-led transformation strategy. Lenovo services handle all required routine maintenance, including firmware updates, as well as other premium services. Customers have access to utilization data, contracts, invoices and service logs through Lenovo's customer portal. TruScale services include managed services, hardware installation, deployment, system health management, maintenance, 24/7 support and hardware removal at the end of the contract. A dedicated customer success manager is assigned to each customer, providing a single point of contact. Lenovo lets its partners add their own service and solution capabilities under TruScale. TruScale is available in 37 countries as of October 2022, and Lenovo plans to add more locations.

NetApp

Product or Portfolio Overview

NetApp Keystone provides pay-for-use STaaS for hybrid cloud environments. NetApp Keystone provides multiple storage service tiers from high performance to archival flexibly supporting storage area network (SAN), network-attached storage (NAS) and object storage requirements. NetApp Keystone is an on-premises service that can include storage services in any of the three largest clouds with the flexibility to transition data for backup, tiering or migration to these clouds as part of the subscription. NetApp Keystone offers STaaS for customers that prefer 100% opex billed monthly based on consumption. NetApp Keystone can be operated by the customer, NetApp or a NetApp partner. It is also available as part of infrastructure-as-a-service offerings delivered jointly with Cisco, NVIDIA, Kyndryl and Equinix. Keystone is available in 16 countries as of September 2022.

This research note is restricted to the personal use of GERMAN.AGUNDEZ@IKUSI.COM.

How This Provider Competes

NetApp Keystone offers comprehensive and flexible consumption models that extend from on-premises to the edge to the cloud for primary storage workloads across its storage arrays and cloud storage services. NetApp delivers AIOps and cloud management tools using BlueXP, which includes Active IQ, and Cloud Insights to maintain service availability and automate actions. NetApp BlueXP orchestrates all tools and provides a single dashboard view to monitor and manage Keystone services remotely, spanning on-premises and the public cloud infrastructure. A single Keystone subscription can include both on-premises storage and cloud storage services (NetApp Cloud Volumes ONTAP and NetApp Cloud Backup), supporting the three major cloud providers.

Pure Storage

Product or Portfolio Overview

Pure Storage recently expanded its subscription offerings and combined all programs under its Evergreen brand. The new portfolio consists of Evergreen//Forever (appliance ownership with a subscription to software and hardware upgrades, formerly Evergreen Gold), Evergreen//Flex (a new site-level subscription that combines hardware ownership with usage-based subscription), and Evergreen//One (SLA and consumption-based service model, formerly Pure as-a-Service). Evergreen//One provides a unified subscription licensing across hybrid cloud environments, as well as customized solutions based on true consumption. It covers block, file and object storage services for on-premises, at colocation/hosting providers, or in the cloud. Pure Storage also partners with Cisco to provide FlashStack as a service, a software-defined hybrid-cloud infrastructure solution that integrates compute, network and storage.

How This Provider Competes

Pure Storage's product is often regarded as simple to install and manage. It has extended this simplicity into its Evergreen subscriptions. Its Evergreen offers a unified data storage subscription under a single contract with common management across on-premises and public cloud deployments regardless of deployment location. Evergreen//Forever subscription is offered at a flat rate based on array models. Evergreen//Flex and Evergreen//One offer per-unit pricing based on your usage or consumption. Its AIOps platform and cloud-based data management tool, Pure1 (included in the subscription) offers proactive support, AI-driven monitoring and analytics, and real-time simulation modeling. The company also offers deep integration at the application and container level. Evergreen subscriptions are available in all of the countries where Pure Storage does business (44 countries worldwide) as of September 2022.

This research note is restricted to the personal use of GERMAN.AGUNDEZ@IKUSI.COM.

References and Methodology

Primary and secondary resources were used to prepare this research. We used additional industry sources and publicly available information to verify the accuracy of the information. Sources of data used by Gartner include the following:

- Technology provider briefings and interviews
- Data from Gartner interaction with end users and technology providers
- Gartner end-user surveys
- Articles in the general and trade press
- Published company announcements and financial earnings reports

In addition, factual review of the technology providers' information was conducted by the respective technology providers. Our conclusions about competitive positioning consider these inputs but, ultimately, reflect Gartner's own judgment based on our overall perspective of the market.

Recommended by the Authors

Some documents may not be available as part of your current Gartner subscription.

Forecast: External Controller-Based Storage, Worldwide, 2021-2027, 4Q23 Update

Forecast Analysis: External Controller-Based Storage, Worldwide

Market Share: Data Center Hardware Integrated Systems, Worldwide, 3Q23 Update

Market Share Analysis: Data Center Hardware Integrated Systems, Worldwide, 2022

This research note is restricted to the personal use of GERMAN.AGUNDEZ@IKUSI.COM.

© 2025 Gartner, Inc. and/or its affiliates. All rights reserved. Gartner is a registered trademark of Gartner, Inc. and its affiliates. This publication may not be reproduced or distributed in any form without Gartner's prior written permission. It consists of the opinions of Gartner's research organization, which should not be construed as statements of fact. While the information contained in this publication has been obtained from sources believed to be reliable, Gartner disclaims all warranties as to the accuracy, completeness or adequacy of such information. Although Gartner research may address legal and financial issues, Gartner does not provide legal or investment advice and its research should not be construed or used as such. Your access and use of this publication are governed by Gartner's Usage Policy. Gartner prides itself on its reputation for independence and objectivity. Its research is produced independently by its research organization without input or influence from any third party. For further information, see "Guiding Principles on Independence and Objectivity." Gartner research may not be used as input into or for the training or development of generative artificial intelligence, machine learning, algorithms, software, or related technologies.

This research note is restricted to the personal use of GERMAN.AGUNDEZ@IKUSI.COM.

Table 1: CBM IT/AI Capabilities by Vendor

Vendor	Console	AIOps (integrated in console) — management, analytics and cloud ops management
Cisco	Cisco Intersight and Cisco Nexus Dashboard	Cisco Intersight Infrastructure Service and Cisco Intersight Workload Optimizer
Dell Technologies	APEX Console	APEX Console, CloudIQ and others
Hewlett Packard Enterprise (HPE)	HPE GreenLake edge-to-cloud platform	HPE InfoSight and Aruba Central
Hitachi Vantara	Hitachi Ops Center	Hitachi proprietary and third-party tools
IBM Power	IBM Cloud Management Console, Hardware Management Console	IBM proprietary tools
IBM Storage	IBM Storage Insights	IBM proprietary tools
Lenovo	XClarity	XClarity, NetApp ONTAP and third party
NetApp	BlueXP	Active IQ and Cloud Insights
Pure Storage	Pure1	Pure1

Source: Gartner (November 2022)

This research note is restricted to the personal use of GERMAN.AGUNDEZ@IKUSI.COM.